COURSE TITLE : COATING POWDERING AND DRYING
COURSE CODE : 5103
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 44
CREDIT :4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Field of Application	12
2	Basic objectives of inline coating	10
3	UV coating and Hybrid	12
4	Press configuration	10
TOTAL		**44**

Course Outcomes:

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To understand field of application for inline finishing process
2	To comprehend objectives of inline coating
3	To know applying process
4	To comprehend press configurations

MODULE I FIELD OF APPLICATION

1.1.0 To understand field of application for inline finishing process

1.1.1. To define Packaging application
1.1.2. To specify Label application
1.1.3. To define Catalogues application
1.1.4. To define Folding boxes appliaction

MODULE II BASIC OBJECTIVES OF INLINE COATING

2.1.0 To comprehend objectives of inline coating

2.1.1 To explain Increase of visual effects, particularly gloss
2.1.2 To explain Protection of the substance from mechanical factors
2.1.3 To specify Various types of coatings
2.1.4 To describe Basic facts about coating and drying
2.1.5 To explain Inline coating unit with conventional system

MODULE III UV COATING AND HYBRID

3.1.0 To know applying process

- 3.1.1 To define Press with fully integrated UV-equipment
- 3.1.2 To explain about model of UV-end-of-press dryer
- 3.1.3 To specify Principles of UV-Coatings
- 3.1.4 To state Aspects of UV-Coating
- 3.1.5 To explain The Hybrid Process

MODULE IV PRESS CONFIGURATION

4.1.0 To comprehend press configurations

- 4.1.1 To explain possible delivery configurations
- 4.1.2 To define Technology features
- 4.1.3 T o explain Technical concept of inter deck dryers
- 4.1.4 To explain Delivery configurations based coating-Drying-Drying-Coating
- 4.1.5 To explain Dryer concept of double coating

CONTENT DETAILS

MODULE I

Widely used inline coating - its application field Packaging, -Labels - Catalogues –Folding boxes – Posters – Prospects and a lot more. Possibilities and potential for printers with inline coating. Cost reduction and gain - Generation of additional value – Getting into new markets - Cost wise alternatives.

MODULE II

Increase of visual effects – Protection of the substrate from-mechanical factors such as abrasion and scratches-penetration by liquids, moisture and gases-finger prints - Possibilities of coating effects – Various types of coatings – Solid - Spot coating – solid with cutout – gold and silver effect - Possible gloss points of inline coatings - Basic facts about coating and drying – Application of print varnish allows a flexible use of the press – Print varnish applied by the printing unit - Drying process of print varnishes - Drying process of aqueous coatings is based on physical drying Contactless pile temperature control - Inline coating unit with conventional system - Different coating systems –features.

MODULE III

Press with fully integrated UV-equipment – Principles UV-Coating on top of conventional ink - UV-Varnish on top of primer Conventional - Aspects of UV-Coating - UV-Coating applied on Hybrid Ink - Gloss effects with Hybrid Technology – Flexokit for coating units – Thickness of pigments -

MODULE - IV

Possible delivery configurations - Technology features - Technical concept of inter deck dryers - Delivery configurations based coating-Drying-Drying-Coating - Dryer concept of double coating- Dryer concept of LYYL-configuration.

Reference :		
Author	**Title**	**Publishers**
Print media Academy	Offset printing inline Coating and Drying	PMA , Heidelberg
GATF	Lithographers manual	GATF , USA
Adams J.M	Printing Technology	Delmar Publishers, NewYork
Helmut Kipphan	Hand book of print media	Springer Science & Business Media